

COURSE TITLE : PAPER BOARD AND SPECIALITY PAPER
COURSE CODE : 5123
COURSE CATEGORY : E
PERIODS PER WEEK : 4
SEMESTER : 5
PERIODS PER SEMESTER : 52
CREDITS : 3

TIME SCHEDULE

Module	Topics	Periods
I	Paper Boards Test	12 1
II	Coating of paper Speciality paper Test	6 6 1
III	Properties and Testing of Paperboards Test	12 1
IV	Printability of Paper Test	12 1
Total		52

Course Outcome :

Module	G.O	Student will be able to:
1	1	Understand Paper Boards
2	1	Understand Coating of Paper
	2	Understand Specialty Papers
3	1	Understand properties of Paper and Paper Boards
4	1	Understand Printability of Paper

Specific Outcome

MODULE– I

1.1.0 Understand Paper Boards

- 1.1.1 Define Paper Boards
- 1.1.2 List Paper Boards
- 1.1.3 Explain manufacturing process
- 1.1.4 Explain constructional details of Multiply boards with figure
- 1.1.5 Explain properties of Board
- 1.1.6 Define Corrugating Medium

- 1.1.7 Explain pulping system of Multiply boards
- 1.1.8 Classify grades of Paper Board
- 1.1.9 Define Electrical Insulating Board
- 1.1.10 Explain Corrugate Adhesives
- 1.1.11 Define Box board
- 1.1.12 Explain Box board.

MODULE – II

2.1.0 Understand Coating of Paper

- 2.1.1 Define Coating
- 2.1.2 Define Air Knife coating
- 2.1.3 Explain with figure
- 2.1.4 Define Brush coating
- 2.1.5 Explain with figure
- 2.1.6 Define Cast coating
- 2.1.7 Explain Cast coating
- 2.1.8 Define Flexible blade coating
- 2.1.9 Explain Gravure coaters
- 2.1.10 Define Knife coater, Reverse roll coater, Transfer roll coater

2.2.0 Understand Specialty Papers

- 2.2.1 Define Specialty paper
- 2.2.2 List Specialty paper
- 2.2.3 Explain Grease Proof paper, Cigarette paper, Tissue paper, Butter paper, Insulatory paper, Filter paper, Newsprint, Sanitary towels.

MODULE- III

3.1.0 Understand properties of Paper and Paper Boards

- 3.1.1 Define paper
- 3.1.2 Define chemical testing of pulp
- 3.1.3 Define physical testing of pulp
- 3.1.4 Explain Mechanical Properties of paper
- 3.1.5 List mechanical properties of paper
- 3.1.6 Explain method of determining physical properties such as Surface smoothness, Porosity, Basis weight as per TAPPI
- 3.1.7 List mechanical properties
- 3.1.8 Explain method of determining Tear, Stretch, Compression, Tension, Folding endurance as per TAPPI
- 3.1.9 Explain Brightness, Opacity, Gloss, Water absorbency etc as per TAPPI

MODULE- IV

4.1.0 Understand Printability of Paper

- 4.1.1 Define Printing
- 4.1.2 Define Printability of paper
- 4.1.3 Explain Letter press
- 4.1.4 Explain Lithography
- 4.1.5 Explain Screen printing
- 4.1.6 Explain Printing Ink
- 4.1.7 Explain Paper requirement for printing
- 4.1.8 Explain Reprography
- 4.1.9 Explain Xerox technology
- 4.1.10 Explain Thermography
- 4.1.11 Explain Gravure Printing

CONTENT DETAILS

MODULE –I:

PAPER BOARDS- Paper board- Introduction- Definition- Manufacturing process- Fiber analysis of paper. Multiply paper board- Definition- Manufacturing process- Cylinder type formers. Multiply board cross section with figure- Multiple Fourdriner board machine with figure- Multiply board propertiesIndustrial grade paper boards- Linear board- Corrugating medium- Construction paper. Liner board- Definition- manufacturing using Fourdriner machine- Pulping system for Liner board- Paper board treatments Grades and types of board- Electrical insulating boards- Stereo type mat- matrix board- Rubber board- Corrugated containers- materials- Combined board structure- Corrugated adhesive- box board.

MODULE- II:

COATING OF PAPER Coating- Coating equipment- Air Knife coater- Brush coater- Cast coating- Flexible blade coater- Gravure coater- Reverse roll coater- Transfer roll coater. SPECIALITY PAPER Grease proof paper- Cigarette paper- Tissue paper- Butter paper- Insulating paper- Filter paper- Newsprint- Sanitary towels

MODULE- III:

PROPERTIES AND TESTING OF PAPER AND PAPER BOARDS Objectives of testing- Chemical testing of pulp- Physical testing of pulp- Testing procedure- Paper testing- physical, mechanical properties- Basis weight- Caliper- Density- Bulk- folding endurance- Tear- Burst- Tensile strength- Stretch- Softness- Roughness. OPTICAL PROPERTIES Brightness- Opacity- Gloss- water absorbency as per TAPPI

MODULE- IV:

PRINTABILITY OF PAPER- Printing- Printing process- Letter- Gravure- Lithography- Screen printing- Inks for letter press and lithography- Paper requirement for Letter press and Lithography- Reprography- Xerox technology- Thermography- Gravure printing.

REFERENCE

1. K. W. Britt, Pulp and Paper Technology, CBS
2. Robert. R.A & Higham ,Paper Board and Board- ,CBS
3. G. A. Smooke, Hand Book of Pulp and Paper Technology- Interscience
4. Pulp and Paper Manufacture. Vol. I, II, III by Ronald G. Macdonald & John. N. Franklin